

Business 4720 - Class 1

Introduction and Processes

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Terminology

- ▶ Data Analytics
- ▶ Data Science
- ▶ Business Analytics
- ▶ Machine Learning
- ▶ Artificial Intelligence
- ▶ Big Data
- ▶ Statistics

Terminology

- ▶ Method
- ▶ Technique
- ▶ Tool

Types of Analytics

Aims and Outcomes

- ▶ *Descriptive Analytics*: Describes "what is". Summarizes the data, makes comparisons, identifies historical trends. Typically model-free.
- ▶ *Predictive Analytics*: Predicts "what may be" in the future. Typically builds a model based on past data to predict future cases/events/outcomes.
- ▶ *Prescriptive Analytics*: Prescribes "what should be done". Builds a model that learns best actions based on past observations and actions.
- ▶ *Visual Analytics*: Uses graphs to visualize information for gaining insight. Employs human abilities to visually identify trends, make comparisons, etc.

Data

- ▶ In many analytics applications, 80% of time/cost/effort is spent on data management
- ▶ Data quality determines the quality of the analytics outcome

Models

- ▶ Statistical models, for example a linear regression model for prediction
- ▶ Describes the data (generating mechanism) in parameterized form, for example intercept and slope of linear model.

Supervised Learning

- ▶ Correct outcome (numerical value or target category) is given for each input/observation
- ▶ Examples: Linear regression model, generative pre-trained transformers (GPT)

Unsupervised Learning

- ▶ Unsupervised learning: No outcomes are provided
- ▶ Example: Clustering, components analysis

Analytics is not Statistics

- ▶ Both may use the same kinds of mathematical models

Explanation

- ▶ Sample or population statistics
- ▶ Model quality determined by fit between model and data
- ▶ Inferential statistics concerned with generalizing from sample to population
- ▶ Aims to identify data generating mechanism
- ▶ Not focused on individual cases/observations

Prediction

- ▶ Focus on individual cases/observations
- ▶ Model quality determined by precision/accuracy of individual predictions
- ▶ No inference to population
- ▶ Pragmatic, does not aim or claim to identify data generating mechanism

What You Will Learn:

- ▶ Introduction to Business Analytics
- ▶ Data Management (On-Premises & Cloud; SQL & NoSQL)
- ▶ Computational Foundations
- ▶ Descriptive and Visual Analytics
- ▶ Supervised and Unsupervised Machine Learning
 - ▶ Regression, Classification
 - ▶ Clustering, Component Analysis
 - ▶ Time-Series Models, incl. ARIMA and GARCH
- ▶ Predictive Analytics with Deep Neural Networks
- ▶ Prescriptive Analytics (Reinforcement Learning)
- ▶ Process Analytics (Mining and Prediction)
- ▶ ML and AI Processes
- ▶ Ethical and Legal Issues
- ▶ Management and Governance

- ▶ Open-source
- ▶ Cross-platform (Linux, Windows Mac)
- ▶ Multi-architecture (x64/amd64 "Intel/AMD" and arm64 "ARM/Apple")



- ▶ Language and Software for Statistical Computing
 - ▶ First appeared August 1993
 - ▶ Open-source, multi-platform
 - ▶ More than 16,000 packages available on CRAN ("Comprehensive R Archive Network")
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- ▶ Tidyverse packages (tidyr, dplyr, stringr)
- ▶ GGPlot2

<https://www.r-project.org/>



- ▶ Programming Language
 - ▶ First appeared February 1991
 - ▶ Open-source reference implementation, multi-platform
 - ▶ More than 450,000 packages available on PyPI ("Python Package Index")
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- ▶ PyCharm
- ▶ JupyterLab Desktop
- ▶ Numpy, Pandas, Plotly
- ▶ Tensorflow, Keras

<https://www.python.org/>



- ▶ Relational Database Management System
 - ▶ Open-source, multi-hardware
 - ▶ First appeared July 1996
 - ▶ Active DBMS features such as triggers and stored procedures in PL/SQL and other languages
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- ▶ Graph Database (NoSQL)
- ▶ Supports property graphs
- ▶ Open-source community edition
- ▶ Cypher query language

Software Versions and Sources

R	4.1.2	www.r-project.org
Python	3.8	www.python.org
Postgres	16.0-1	www.postgresql.org
Neo4J	5.14.0	www.neo4j.com

R Libraries and Versions

ggplot2	3.4.4	www.tidyverse.org
scales	1.3.0	scales.r-lib.org/
dplyr	1.1.3	www.tidyverse.org
tidyr	1.3.0	www.tidyverse.org
readr	2.1.4	www.tidyverse.org
lubridate	1.9.3	www.tidyverse.org
ggradar	0.2	github.com/ricardo-bion/ggradar
ggpattern	1.0.1	cran.r-project.org/package=ggpattern
ggsci	3.0.0	nanx.me/ggsci/
sqldf	0.4-11	cran.r-project.org/package=sqldf
ISLR2	1.3-2	cran.rstudio.com/package=ISLR2
glmnet	4.1-8	glmnet.stanford.edu
class	7.3-20	cran.r-project.org/package=class
ROCR	1.0-11	cran.r-project.org/package=ROCR
e1071	1.7-14	cran.r-project.org/package=e1071
astsa	2.1	cran.r-project.org/package=astsa
zoo	1.8-12	cran.r-project.org/package=zoo
fGarch	4032.91	cran.r-project.org/package=fGarch

Python Packages and Versions

numpy	1.26.2	numpy.org
pandas	2.1.2	pandas.pydata.org
plotly	5.18.0	plotly.com
kaleido	0.2.1	github.com/plotly/Kaleido
pm4py	2.7.8.4	processintelligence.solutions/p
tensorflow	2.14.0	www.tensorflow.org
tensorflowjs	4.20.0	www.tensorflow.org/js
tensorflow_datasets	4.9.9	www.tensorflow.org/datasets
keras	2.14.0	keras.io
scikit-learn	1.3.2	scikit-learn.org
imblearn	0.13.0	imbalanced-learn.org
statsmodels	0.14.1	www.statsmodels.org
gymnasium	0.29.1	gymnasium.farama.org
pygame	2.5.2	www.pygame.org
shap	0.45.0	github.com/shap/shap
lime	0.2.0.1	github.com/marcotcr/lime
PyALE	1.2.0	pypi.org/project/PyALE/
pyspark	3.5.1	spark.apache.org
Werkzeug	2.3.8	werkzeug.palletsprojects.com
Flask	2.2.3	flask.palletsprojects.com

Database Management Systems

- ▶ MongoDB (a document database)
- ▶ ArangoDB (a multimodel database)
- ▶ OrientDB (a graph database)
- ▶ Redis (a key/value database)
- ▶ Cassandra (a NoSQL database)

Infrastructure

- ▶ Hadoop (a distributed file system)
- ▶ Spark (an analytics engine on Hadoop)
- ▶ HBase (a distributed database)
- ▶ Hive (a data warehouse system)

Download and install the required software on your computer.

- ▶ R → www.r-project.org
- ▶ Python → www.python.org
- ▶ Postgres → www.postgresql.org
- ▶ Neo4J → www.neo4j.com